COUNTRY PROFILES

# Nuclear Power in Iran

UPDATED MONDAY, 23 MARCH 2026

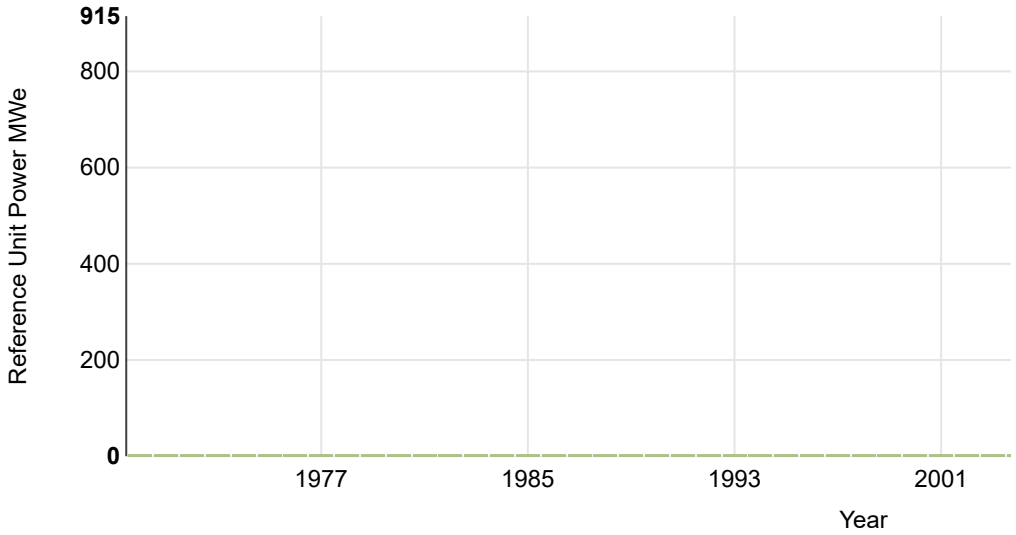
- One nuclear power reactor is operating in Iran and another is under construction.
- Construction commenced on a second large Russian-designed unit at the Bushehr site in November 2019, and a third unit is planned.
- The country also had a major programme developing uranium enrichment, which it concealed for many years.
- Iran began limiting its enrichment-related activities and ceasing its work on heavy water-related projects under the internationally-agreed Joint Comprehensive Plan of Action (JCPOA). However, following the USA's withdrawal from the agreement in 2018, the country began enriching uranium again.
- On 12 June 2025 the International Atomic Energy Agency formally declared Iran in breach of its non-proliferation obligations. The following day, Israel launched a military operation targeting Iran's military and nuclear fuel cycle sites, substantially damaging them (see below).
- In October 2025 Iran officially ended the 2015 JCPOA, declaring that all restrictions on its nuclear programme are now void.

Privacy - Terms

On 1 March 2026 the USA and Israel began a joint military campaign which struck some nuclear fuel cycle sites once again (see below).

| 1 | 1 | 0 |
|---|---|---|
| OPERABLE REACTORS | REACTORS UNDER CONSTRUCTION | REACTORS SHUTDOWN |
| 915 MWe | 974 MWe | 0 MWe |

## Operable nuclear power capacity




# Electricity sector

**Total generation (in 2023):** 385 TWh

**Generation mix:** natural gas 303 TWh (79%); oil 57.1 TWh (15%); hydro 16.9 TWh (4%); nuclear 6.0 TWh (2%); solar 0.9 TWh; wind 0.8 TWh; coal 0.7 TWh.

**Import/export balance:** 0.9 TWh net export (4.0 TWh imports; 4.9 TWh exports)

**Total consumption:** c. 332 TWh

**Per capita consumption:** c. 3700 kWh in 2023

*Source: International Energy Agency and The World Bank. Data for year 2023.*

Demand for electricity is growing at about 4% per year, and Iran trades electricity with Afghanistan, Armenia, Azerbaijan, Iraq, Pakistan, Syria, Turkmenistan and Turkey. At the end of 2024 total generating capacity was about 90 GWe.

# Nuclear power industry

| Reactor Name | Model | Reactor Type | Net Capacity (MWe) | Construction Start | First Grid Connection |
|---|---|---|---|---|---|
| Bushehr 1 | VVER V-446 | PWR | 915 | 1975-05 | 2011-09 |

## Nuclear industry development

In 1957 a civil nuclear programme was established under the US Atoms for Peace programme.

In 1974 the Shah announced a target of 23,000 MWe of nuclear capacity to free up oil and gas for export. Preliminary agreements with Siemens KWU and Framatome for four nuclear power plants were signed.

In 1975 construction of two 1293 MWe (gross) PWR units was started 18 km south of **Bushehr** in Bushehr province on the Persian Gulf by Siemens KWU, based on the Biblis B reactor in Germany. The contract was actually signed in mid-1976 and some $3 billion paid. After the Islamic revolution, further payment was withheld and work was abandoned early in 1979 with unit 1 substantially complete and unit two about half complete. The site was damaged by Iraqi air strikes in 1984-88.

At **Darkhovin**, on the Karun River close to the Iraq border, there were also two French 910 MWe units which in January 1979 had just started construction under a $2 billion October 1977 contract with Framatome. These were cancelled in April 1979, and their engineering components were retained in France, being built there as Gravelines C, units 5&6, which came online in 1985. In 1992, the Islamic Republic of Iran signed an agreement with China to build two 300 MWe reactors at the Darkhovin site, similar to those at Qinshan in China and Chashma in Pakistan, but China withdrew before construction started.

The original 1974 plan called for construction of four units at Bushehr, then two units at Isfahan, 340 km south of Tehran, to come online in the mid-1980s and two units at Saveh, near Tehran. The Isfahan and Saveh units were to be 1300 MWe class KWU types with dry cooling using two 260 m tall and 170 m wide dry cooling towers. They would have been the first large nuclear plants to use dry cooling.

In November 2025 Mohammad Eslami, head of the Atomic Energy Organization of Iran (AEOI), said that locations for eight new units as part of a programme to build 20 GWe of new nuclear capacity under a partnership with Russia would soon be confirmed. He added that a potential new site had been identified in Golestan province on Iran's northern coast, as well as the possible return to construction of the part-built pre-1979 plant in Darkhovin.

## Bushehr 1

An agreement between the Iranian and Russian governments on building a two-unit nuclear power plant was signed in Moscow in August 1992. This covered both construction and operation of the plant. The Atomic Energy Organization of Iran (AEOI) insisted that the project should make full use of the structures and equipment already at Bushehr. In 1994, Russia's Minatom agreed with the AEOI to complete unit 1 of Bushehr nuclear power plant with a VVER-1000 unit, using mostly the infrastructure already in place, and a contract was signed in January 1995.

### Construction

The Russian contractor faced major challenges and an approach to Germany for help was rejected, leaving it to certify and document a lot of unfamiliar pieces of equipment. All the main reactor components were fabricated in Russia under a construction contract with Atomstroyexport, based on the V-392 design, but designated V-446 signifying adaptation to the Siemens parts and also high seismic rating. The 1000 MWe (915 MWe net) plant constructed by Atomstroyexport had a succession of construction and start-up delays, and as late as 2007 the project was almost abandoned. All work was under IAEA safeguards and operation is also under safeguards. Some 47,000 pieces of equipment passed scrutiny by Atomstroyexport, another 11,000 seemed to be in working order but the specifications and manuals to them were missing and needed to be restored. Meanwhile, since the 1975

construction start, the nuclear safety requirements in Russia and internationally had become more stringent. Some German parts being integrated into the Russian design therefore required significant upgrades. According to a local report, 24% of the parts are German in origin, 36% are Iranian-made while 40% are Russian-made.

The original plans had two desalination plants, each of 100,000 $m^3$/day capacity, linked to the reactors, but the Russian project dropped these. It appears that they were added later, and the first phase was commissioned in August 2014. With phase 2, the plant will provide 20% of the potable water for Bushehr city.

After two years delay due to Iran's reluctance to agree to returning used fuel to Russia without being paid for it, two agreements were signed early in 2005 covering both supply of fresh fuel for the new Bushehr nuclear reactor and its return to Russia after use. The Russian agreement means that Iran's nuclear fuel supply is secured for the foreseeable future, removing any justification for enrichment locally.

The reactor was finally due to start up in February 2011 and to "reach first power" in April, and fuel had been loaded by the start of December. However, late in February before starting up it was found that a pump had failed and possibly shed metal particles into the primary cooling system, which could damage the fuel elements. The fuel therefore had to be unloaded, checked and cleaned, and any debris removed from the pressure vessel. The pump concerned was one supplied in the 1970s and was part of the old equipment required to be used under the terms of the Russian contract. Eventually the reactor started up on 8 May 2011, was grid connected early in September 2011, and was expected to enter commercial operation about April 2012, then May 2013, and finally did so in September 2013.*

* In October 2012 the fuel assemblies were unloaded and put into the spent fuel pond after only two months irradiation, apparently due to indications of some metal fragments in the cooling water remaining from the earlier pump problem. The reactor was reloaded and restarted in December. However, apparently the IAEA was not fully informed of the situation, which caused some international concern about what was happening and why. It was reconnected to the grid in January 2013 and handed over to Iran in September 2013, making it effectively in commercial operation.

*President Hassan Rouhani and Ali Salehi, head of the Atomic Energy Organization of Iran at Bushehr in 2016 (IRNA)*

## Operation

The plant remained under Russian guarantee for another two years from September 2013 with a number of Russian experts to give advice and technical assistance.* The preliminary agreement stated that the first reactor of Bushehr nuclear power plant would be operated by a 50-50 Russian-Iranian joint venture during the one-year warranty period. In August 2010 it was agreed that this JV would operate the plant for up to three years before Atomstroyexport gradually withdrew. However, in September 2011 after grid connection, Rosatom said: "According to the Iranian legislation, only a national company can be an operator of the nuclear power plant", hence Russian specialists would be invited to work under a contract to be awarded by the Nuclear Power Production and Development Company of Iran (NPPD), with their numbers gradually reducing. In May 2012 the first deputy director general of Rosenergoatom Vladimir Asmolov said that all operations related to the reactor equipment control and operation were being carried out by Russian specialists. “There is not a single Iranian operator at Bushehr,” he said. He explained that the Iranian party had signed an agreement with a Rosatom affiliate,

Atomtechexport (ATE), which operates the reactor unit. However, Iranian operators have been trained in Russia.

* A Russian report in January 2014 said that 270 employees of Atomtechexport were insured to work in Iran to support "organization of Bushehr nuclear power plant operation, repair and render engineering support to the facility, including the equipment integration."

The plant is designed to withstand magnitude 8 earthquakes. However, in April 2013 an earthquake of 7.7 magnitude, followed by a smaller tremor in May, reportedly caused cracks in concrete at Bushehr, but without safety significance. Iran reported to the IAEA that the earthquakes had caused no damage. In June 2013 the reactor remained shutdown due to generator problems, and had apparently been out of action for several months. Following the earthquake, the Gulf Cooperation Council met and expressed concern about possible radiation releases from the Bushehr nuclear plant. The six GCC Arab states had previously urged Tehran to ensure its facility complies with international safety standards and to join the IAEA Convention on Nuclear Safety.

The 6-7 TWh/yr from the Bushehr reactor frees up about 1.6 million tonnes of oil (11 million barrels) or 1800 million cubic metres of gas per year, which it intended to export for hard currency, although subsequent sanctions prevented this. In 2019 the AEOI put the annual saving of oil at $660 million per year. By the end of 2019 the reactor had produced an estimated 37 TWh.

Despite high-profile and serious disagreements with the IAEA over uranium enrichment, the IAEA continues full involvement with Iran on nuclear safety issues, focused on Bushehr. However, Iran is not a party to the IAEA Convention of Nuclear Safety – it is the only country with an operating nuclear reactor that is not a party to it.

## Further nuclear power capacity: Bushehr phase II

Through to February 2014 a series of confusing and sometimes conflicting announcements from AEOI showed the intention for further Russian capacity at Bushehr. This would not be integrated into the German-built structure for unit 2.

In March 2014 AEOI said it had agreed with Rosatom to build at least two more 1000 MWe nuclear reactors at Bushehr. AEOI said that the agreement is part of a 1992 deal between the two countries on further nuclear cooperation. Technical and commercial

issues were to be worked out, but Iran's ambassador in Moscow earlier said that the plant, along with other goods, would be bartered for oil (which was then subject to UN trade sanctions). A government-level protocol on building two further reactors was signed in April 2014.

In November 2014 a further protocol to the original 1992 agreement was signed by Rosatom and AEOI, covering construction of four VVER reactors on a turnkey basis at Bushehr, and four more at another site yet to be determined. These are all to involve maximum local engineering content, and will be fully under IAEA safeguards. As usual with its foreign projects, Rosatom would supply all the fabricated nuclear fuel for the eight units "for the whole period of the nuclear power plant operation" and would take all used fuel back to Russia for reprocessing and storage. (Article 3 of protocol.) However, under the terms of the 1992 agreement, Rosatom and AEOI also signed a memorandum of understanding to "work on necessary arrangements for the fabrication in Iran of the nuclear fuel or its elements to be used in Russian design units."

Also in November 2014 a contract for construction of the first two reactors as Bushehr phase II was signed by NIAEP-ASE and the Nuclear Power Production and Development Company of Iran (NPPD). Two desalination plants are to be part of the project. Rosatom said they would be paid for progressively by Iran in the same way as with unit 1. Site works were expected to start by March 2016, but the AEOI called for delay due to technical issues, in particular agreement on seismic parameters. A foundation stone was formally laid in September 2016 with some fanfare to inaugurate the project, and the AEOI said that it would take ten years to build and cost $10 billion. Rosatom said the reactors were to be AES-92 Generation III+, apparently V-466B, based on the VVER-1000/V-392 reactor. In November 2016 the AEOI said that construction would be speeded up. Atomenergoproekt had a contract for detailed design of the two units by August 2018. Iran is to finance them. Site works for construction of unit 2 started in March 2017, "initial excavation" was in October 2017, and first concrete was poured in November 2019. Initial excavation works were under way on unit 3 in January 2021.

The AEOI said in May 2019 that the two reactors would avoid the use of 22 million barrels of oil per year.

In August 2022 work commenced on the country's first nuclear-powered desalination plant (located at Bushehr), with an installed capacity of 70,000 $m^3$/d.

In January 2026 third tier containment was installed at Bushehr 2. As of February 2026, the combined Bushehr phase II project (units 2&3) was reported to be 18% complete, with commercial operation of unit 2 then expected in 2029.

## Nuclear power reactors under construction

| | Type | MWe gross | Construction start | Commercial operation |
|---|---|---|---|---|
| Bushehr 2 | AES-92, VVER-1000/V-466B | 1057 | November 2019 | 2029 |

## Further nuclear power capacity: other sites

### Darkhovin/Darkhovain

In May 2007 the AEOI said it was planning to build an indigenous 360 MWe light water reactor at Darkhowin/Darkhovain on the Karun River in Khuzestan province in the west, close to Iraq at the head of the Gulf. Two Framatome 950 MWe units were about to be built here in 1970s, and two 300 MWe Chinese units were planned in the 1990s. The head of the NPPD denied that these new ones would use Chinese technology and in October 2008 announced that their design would begin shortly and be completed in six years.

In 2011, the AEOI said that it planned a 360 MWe plant in Darkhowin and that its design was well under way, and in May 2012, the AEOI said the design of the light water reactor was finished. A further announcement in February 2013 confirmed planning intention for a 360 MWe plant. In May 2013, a senior government official said that Iranian experts were designing a light water reactor for Darkhowin, under IAEA supervision. The IAEA requested design information on this proposed plant but said in 2013 that it had not been provided. In May 2014 the AEOI said it had made progress on the project. In December 2022 the AEOI announced that “construction” of the unit was launched. It said that the project would take an estimated eight years and cost $1.5-2 billion to complete.

In December 2022 the AEOI announced that groundworks had commenced at Darkhovin; and in October 2023 the AEOI held a

ceremony at the site to commemorate the start of site development works. As of March 2026, construction had not officially started.

The International Atomic Energy Agency (IAEA) requested in February 2024 that Iran provided design information in accordance with its safeguards agreement for the planned IR-360 reactor, as well as preliminary design information for the Hormoz nuclear plant (see below).

## Hormoz nuclear power plant

In February 2024 the AEOI announced the start of civil construction work on the Iran-Hormoz plant near the cities of Minab and Sirik in the southern coastal province of Hormozgan. The total investment of the government's “super project” was put at $15 billion.

In September 2025 Iran signed a $25 billion contract with Rosatom to construct four reactors at the site.

### Other proposed developments

A February 2013 announcement said that 16 sites had been selected for new nuclear plants to be built over the next 15 years. In December 2013 the AEOI said that a majority of Iran’s new nuclear facilities will be on its southern coast on the Persian Gulf and on the northern coast on the Caspian Sea, while another plant would be in central Iran. It was in talks with Rosatom regarding 4000 MWe of new plant, mainly at Bushehr or in Bushehr province. Increasingly, part of the rationale for sites on the Gulf is desalination (for ‘sweet water’), giving them priority in planning.

In July 2015 AEOI announced that China would build two of the next four nuclear power reactors, at a site on the southeastern Makran coast on the Gulf of Oman. Press reports subsequently quoted the AEOI and the vice president that these would be two units of 100 MW each, evidently ACP100 from CNNC, with China having indicated readiness to finance them.

In February 2016 the AEOI agreed to a project with Hungary to design and develop a 25 MWe reactor and another reactor of up to 100 MWe, which could be sold across Asia and Africa while being built in Iran. Meanwhile, some 1100 Iranian students were studying at Hungarian universities, with some emphasis on nuclear technology.

In September 2025, Iran signed a memorandum of understanding with Rosatom for the construction of Russian small modular reactors.

**Nuclear power reactors planned and proposed**

| | Type | MWe gross | Construction start | Commercial operation |
|---|---|---|---|---|
| Bushehr 3 | VVER-1000/V-466B | 1057 | | |
| Darkhovin | IR-360 | 360 | | |
| Total (2) | | 1417 | | |
| Makran coast | ACP100 | 2 x 100 | | |
| Hormoz/Sirik 1-4 | VVER-1200 | 4 x 1250 | | |
| Proposed (6) | | 5200 | | |

# Uranium enrichment

Iran has a major project developing uranium enrichment capability. This programme is heavily censured by the UN, since no commercial purpose is evident. The 2015 Joint Comprehensive Plan of Action (JCPOA) aimed to address this situation – see later section. However, in October 2025 Iran officially ended the JCPOA.

The IAEA estimated that as of 13 June 2025 Iran's total enriched uranium stockpile in the form of $UF_6$ comprised:

- 2391.1 kg of uranium enriched up to 2% U-235.
- 6024.4 kg of uranium enriched up to 5% U-235.
- 184.1 kg of uranium enriched up to 20% U-235.
- 440.9 kg of uranium enriched up to 60% U-235.

## History

The antecedents of this go back to 1974, when Iran loaned $1.18 billion to the French Atomic Energy Commission to build the multinational Eurodif enrichment plant at Tricastin, and it secured a

10% equity in the enterprise, entitling it to 10% of output. The loan was repaid with interest in 1991 but the plant has never delivered any enriched uranium to Iran since the new government in 1979 cancelled its agreements with Eurodif and ceased payments to it. However, in 1991 Iran revived its nuclear power ambitions and demanded delivery of its share of uranium under the original contract, but this was refused by France due to the enforcement of existing political sanctions. Iran viewed this refusal as evidence of the unreliability of outside nuclear supplies and cited the Eurodif experience as the basis for achieving energy independence – through the inhouse development of all of the elements of the nuclear fuel cycle. Eurodif's 10.8 million SWU plant operated by Areva started production in 1979 and closed in mid-2012 – Iran has no equity in its successor.

In about 2000 Iran started building at Natanz, 80 km southeast of Qom, a sophisticated enrichment plant, which it declared to the IAEA only after it was identified in 2002 by a dissident group. This is known as the Pilot Fuel Enrichment Plant (PFEP), and is above ground, but also at Natanz the large underground Fuel Enrichment Plant (FEP) was developed. Operations at the PFEP, FEP and the uranium conversion plant (UCF) are under international safeguards, though monitoring is constrained. In May 2010, environmental samples confirmed that both enrichment plants were operating as declared, with FEP producing less than 5.0% enrichment. However, in February 2010 about 1950 kg of low-enriched uranium from FEP was taken to PFEP. In 2009 the underground Fordow Fuel Enrichment Plant (FFEP) was declared before it became operational.

Over 2009-10 the Iranian centrifuge programme was set back by the Stuxnet computer virus which affected Iranian companies involved with the control systems for the IR-1 centrifuges. In late 2009 to early 2010 about 1000 centrifuges at FEP were decommissioned. This appears to have been due to Stuxnet affecting frequency converters and causing the motors to over-speed, destroying the units. The normal failure rate of the IR-1 centrifuges is reported as about 10% per year.

International concern regarding the surge of activity in enrichment to about 20% U-235 is based on the fact that, in terms of SWU (energy) input, this is about 90% of the way to weapons-grade material, and thus would require only a small and possibly clandestine plant to bridge the gap.

## Joint Comprehensive Plan of Action

On 14 July 2015, China, France, Germany, Russia, UK, USA, the European Union and Iran agreed on the JCPOA.

In May 2018, US President Donald Trump withdrew from the JCPOA.

Between January 2016 and May 2019, the IAEA was able to monitor Iran's nuclear activities and verify they were in accordance with its nuclear-related commitments under the JCPOA. From May 2019 onwards, the Agency said that Iran stopped implementing its commitments on a step-by-step basis, until February 2021, when it stopped implementing them altogether.

In June 2022 Iran removed all of the IAEA's JCPOA-related surveillance and monitoring equipment.

In May 2025 the IAEA said: "As a result of not having been able to perform JCPOA-related verification and monitoring activities for more than four years, the Agency has lost continuity of knowledge in relation to the production and current inventory of centrifuges, rotors and bellows, heavy water and UOC, which it will not be possible to restore."

On 12 June the IAEA's board of governors formally declared Iran in breach of its non-proliferation obligations for the first time in 20 years.

In October 2025 Iran officially ended the JCPOA, declaring that all restrictions on its nuclear programme were void, though Tehran said it remained open to diplomacy.

On 20 November 2025 the IAEA Board of Governors adopted a resolution (19 votes to 3, with 12 abstentions) calling on Iran to cooperate urgently with the Agency. Iran had agreed a verification arrangement with the IAEA in September 2025 (the 'Cairo Agreement') but terminated it following the November resolution. Separately, the E3 (France, Germany, UK) triggered the UN 'snapback' mechanism on 28 August 2025, leading to the reimposition of all previous UN Security Council sanctions on Iran from 28 September 2025.

## Natanz enrichment facility

Natanz is Iran's main uranium enrichment site. It comprises two facilities: the underground Fuel Enrichment Plant (FEP) and the above-ground Pilot Fuel Enrichment Plant (PFEP) (see below). In October 2020 the IAEA released satellite images showing Iran had begun construction of an underground plant at Natanz.

In July 2020, a major explosion occurred at Natanz, damaging a warehouse used to construct centrifuges. The AEOI did not confirm the cause of the explosion "because of security concerns."

**PFEP**

The Pilot Fuel Enrichment Plant (PFEP) is an above-ground facility at Natanz which started operating in 2003.

In May 2010, Iran agreed to enhanced surveillance and checks at the PFEP. On 23 June 2011 the head of AEOI said: "We have the ability to produce 5 kg (of 20% enriched uranium) each month, but we do not rush." He had earlier said that the Tehran Research Reactor (TRR) required 1.5 kg of fuel per month. In August 2011, he confirmed that Iran had more 20% LEU than it needed for the TRR and that "security measures required that the sensitive part of the facilities would be transferred to underground buildings" – evidently Fordow. The IAEA reported then that monthly production rates of 20% LEU had increased significantly, implying better performance of the two IR-1 cascades then in operation. The near 20% $UF_6$ was then being converted to oxide form.

In total, Iran fed 1631 kg of 3.5% LEU to produce 202 kg of 19.75% uranium at the PFEP from the beginning of operation in February 2010 to October 2015. Production then ceased, as agreed under the JCPOA.

In January 2017 the AEOI reported that $UF_6$ was being injected into the first IR-8 centrifuges, within the framework of the JCPOA. It said that these are about 20 times the capacity of IR-1 centrifuges.

In March 2021 Iran restarted enriching uranium at the Natanz facility.

The PFEP is being used to enrich uranium to 60% U-235. The IAEA estimated that between 8 February 2025 and 17 May 2025 19.2 kg of uranium enriched up to 60%, 243 kg up to 5% and 453 kg up to 2% U-235 was produced at the PFEP.

**Natanz FEP**

The main underground Fuel Enrichment Plant (FEP) at Natanz produces uranium enriched to 5% U-235. According to the IAEA a

total of 101 cascades of centrifuges are installed, of which 82 were operating in May 2025. From 8 February 2025 to 16 May 2025 the IAEA estimated that 2671.3 kg of $UF_6$ enriched up to 5% U-235 was produced at the FEP.

## Fordow FFEP

In September 2009, after the fact was exposed internationally, Iran told the IAEA that it was building another enrichment plant at Fordow, about 20 km north of Qom, in an underground tunnel complex on a military base. According to the Iranian 'Nuclear Archive' this was the Al Ghadir project being built by the military from about 2002 originally to produce weapons-grade uranium from LEU supplied by AEOI. As of 2009 this Fordow Fuel Enrichment Plant (FFEP) had been transferred to the AEOI.

In January 2021 the IAEA reported that the Fordow facility was being used to enrich uranium to 20%.

According to the IAEA a total of 16 cascades of centrifuges are installed, of which 13 were operating in May 2025.

Earlier in December 2024 the IAEA reported that Iran had started feeding two IR-6 cascades producing $UF_6$ enriched up to 60% U-235 at the FFEP with $UF_6$ enriched up to 20% U-235, rather than $UF_6$ enriched up to 5% U-235, without altering the enrichment level of the product. The effect of this change has been to significantly increase the rate of production of $UF_6$ enriched up to 60% at the FFEP to over 34 kg of uranium in the form of $UF_6$ per month.

The IAEA estimated that from 8 February 2025 to 16 May 2025 166.0 kg of uranium enriched up to 60%, 724.5 kg up to 5% and 368.7 kg up to 2% U-235 was produced at the FFEP.

## Lashkar Ab'ad laser laboratories

This was the site of experiments on undeclared laser enrichment about 2003, and the facility has expanded greatly in recent years. It is not clear what activities are being pursued there, as IAEA has been denied information and access. However, there are some indications that work on laser enrichment may continue. In 2010 there was a high-profile announcement that the country has laser enrichment capability.

### Isfahan enrichment facility

In June 2025 Iran declared a new underground enrichment facility at the Isfahan nuclear complex to the IAEA. As of March 2026 the IAEA has not inspected the site and does not know whether centrifuges have been installed.

## Conversion and fuel fabrication

A uranium conversion plant (UCF) at the Isfahan Nuclear Technology Centre has 200 t/yr capacity and started up in 2004. It is under IAEA safeguards and the IAEA reported that to November 2014 it had produced 550 tonnes of natural $UF_6$, of which 163 tonnes had been moved to Natanz FEP.

Also at Isfahan, the Enriched $UO_2$ Powder Plant (EUPP) was commissioned in 2014 to convert $UF_6$ to $UO_2$. To November 2015 it had produced 2330 kgU as enriched $UO_2$ product.

A fuel fabrication plant is next to the UCF, and has been operational since 2012. This has continued producing fuel for the Tehran Research Reactor (TRR), and 22% of the 228 kg of near 20% LEU hexafluoride sent for conversion to oxide, has been made into 39 TRR fuel assemblies, containing 51.5 kgU.

In November 2015 Iran no longer had any near 20% $UF_6$, it did have 228 kgU as $U_3O_8$ enriched to that level. It has also started to recover uranium from scrap, and has recovered 44.7 kgU as oxide, with 30 kg of this being used for TRR fuel fabrication.

## Research & development

The Teheran Nuclear Research Centre was established in 1967 by the AEOI. It has a US-supplied 5 MW pool-type research reactor which has operated since about 1967. The IAEA monitors the **Tehran Research Reactor** (TRR) and also a Molybdenum, Iodine and Xenon Radioisotope production facility (MIX). Since being converted from 93% HEU about 1988 by Argentinian specialists, the TRR runs on 19.75% enriched uranium, and 116 kg of this was supplied from Argentina about 1993 – enough for 10-20 years depending on how the reactor is operated. This had nearly run out in 2009. It was earlier said that the presence of molybdenum in

Iranian $UF_6$ means that domestic supplies may be unsuitable at this level of enrichment, but this is unconfirmed.

In 2009 it seemed likely that Russia might provide some further uranium for TRR fuel blended down from 36% enriched material and fabricated in France, in exchange for an equivalent amount of Iran's (< 5%) enriched uranium from Natanz. This was rejected by Iran, which then tabled a revised version. At issue was the amount of Iran's uranium stockpile to be handed over at one time, and where this would occur. The international negotiators wanted to do this exchange in one large shipment, while Iran preferred several smaller swaps which maintained more of its overall holding for a longer period. In February 2010 the government ordered the AEOI to commence enriching Iranian uranium to 19.75%.

In May 2010, a deal with Brazil and Turkey was announced and submitted to the IAEA whereby Iran would ship 1200 kg of 3.5% enriched uranium to Turkey, and then receive 120 kg of 19.75% enriched uranium fuel elements for the TRR in return from the so-called Vienna Group, comprising the IAEA, USA, Russia and France. This apparently did not proceed.

The Nuclear Technology Center of Isfahan operates four small nuclear research reactors, all supplied by China, including a 30 kW MNSR. The centre is run by the AEOI. In April 2013 the AEOI announced plans for a new research reactor at Bonab, and in June the location was reported as Zarghan, in East Azerbaijan province, respectively south and north of Tabriz.

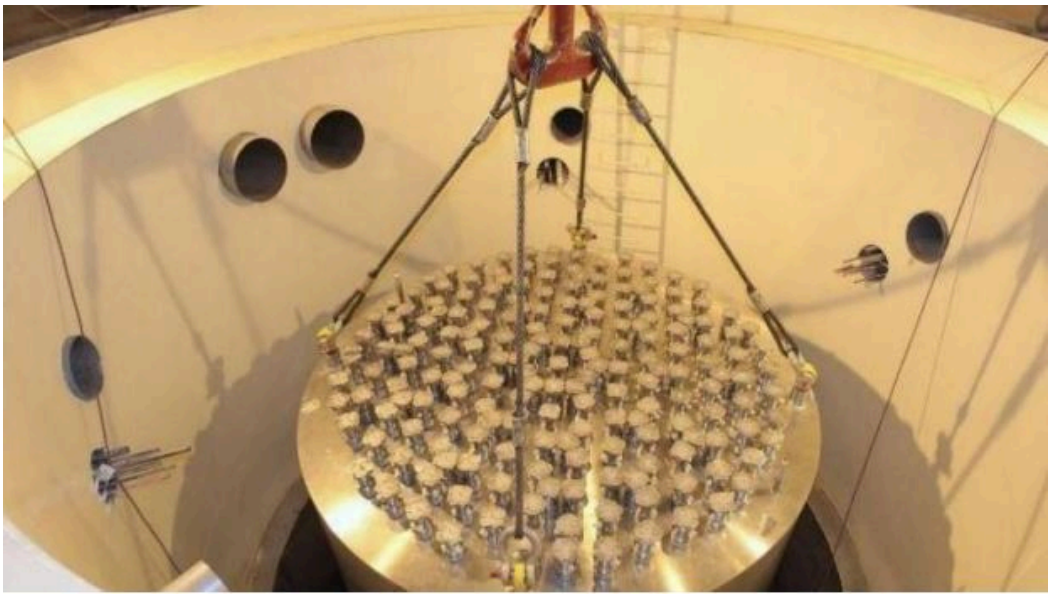
In June 2010, the AEOI announced that it planned to build four new research reactors for production of medical isotopes, including a 20 MW one to replace TRR when its operational life finishes in 15 years. This plan would justify production of more 20%-enriched uranium at Natanz, which gives rise to international concern. In May 2014 the AEOI announced plans for a 10 MW light water research reactor at Shiraz to produce medical and industrial radioisotopes.

## Arak IR-40 heavy water reactor

Iran has also been building a 40 MWt heavy water-moderated reactor at Arak fuelled by natural uranium. The IR-40 reactor is officially referred to as the Khondab Heavy Water Research Reactor. It was declared as a replacement for TRR and the production of radioisotopes.

Apparently designed by Russia's NIKIET, the IR-40 design is very similar to those used by India and Israel to make plutonium for nuclear weapons. The incomplete plant was 'inaugurated' in August 2006. The reactor vessel was due to be installed in 2011. In July 2011 the AEOI reported it as 75% complete. Iran has said that it will be under IAEA safeguards and has been subject to IAEA inspection during construction. However, from 2006 Iran has declined to provide the IAEA with detailed design information on the IR-40 reactor to allay concerns regarding its precise purpose. An IAEA design verification visit in February 2013 noted that cooling and moderator plumbing was almost complete, though in November, critical components were not yet installed. In April 2014, the AEOI announced that an oxygen-18 production unit for positron emission tomography (PET) had started operation at Arak. It was expected to start up in 2014, with a delay due to slow progress with fuel fabrication. However, in October 2014 it was still incomplete, and the reactor has not achieved criticality.

The UN Security Council has demanded that construction of IR-40 cease due to its plutonium production potential. In February 2014 the AEOI said that the reactor was not primarily for plutonium production, that its 9 kg/yr of plutonium would not be weapons grade, and that regardless, the AEOI might redesign it to meet Western concerns. Also, it said that Iran has no reprocessing capacity. US sources have suggested changing the fuel and lowering the power to reduce plutonium production to about 1 kg per year, while still enabling its purported use for making radioisotopes. Following UN acceptance of the Joint Comprehensive Plan of Action in July 2015, the Chinese Foreign Minister announced that China had undertaken a process to modify the Arak reactor and that to effect this, “a joint working group consisting of the six parties and co-chaired by China and the United States will be set up.” In April 2017 Iran signed an agreement with China to modify the reactor.



*Original calandria of IR-40 (ISIS)*

A heavy water production plant is operating at Arak, but the IAEA was denied access to it from 2011 to November 2013. It began construction in 2001, with the capacity to produce 25 tonnes of reactor-grade heavy water per year. About 124 tonnes is stored in Iran; 11 tonnes elsewhere.

A fuel manufacturing plant has been constructed at Isfahan to serve the IR-40 reactor and potentially Bushehr and TRR. In May 2009, the IAEA noted that fuel rods were being produced and that an initial fuel assembly for IR-40 had been produced from these. Production of natural uranium oxide fuel pellets has continued. In November 2012, the IAEA noted that a prototype IR-40 natural uranium fuel assembly was to be irradiation tested at TRR. To May 2015, 36 prototype and 11 final natural uranium fuel assemblies for IR-40 had been produced here, totalling 102 tonnes. Two fuel assemblies using 3.4% enriched uranium of mass 6 tonnes had also been produced. Fuel production stopped in 2014 under the terms of the Joint Plan of Action (see below). The April 2015 interim agreement following this specified that the Arak reactor will be redesigned and its original core, capable of producing significant quantities of weapons-grade plutonium, will be removed and destroyed. No other heavy water reactor is to be built for 15 years. The core was removed in January 2016.

On 19 June the Israeli military bombed the unfinished Arak reactor (see above).

## Israeli military operation, June 2025

On 13 June 2025 Israel launched a military operation targeting Iran's nuclear fuel cycle and military sites. Prime minister Benjamin Netanyahu stated: "If not stopped, Iran could produce a nuclear weapon in a very short time." The Israeli military said that its strikes were focused on "over 100 targets, including senior figures of the Iranian General Staff and leaders of the nuclear program."

The IAEA said on the same day that the above-ground part of the Natanz Pilot Fuel Enrichment Plant, where Iran was producing uranium enriched up to 60% U-235, had been destroyed. The Agency initially said that there was no indication of a physical attack on the underground cascade hall containing part of the Pilot Fuel Enrichment Plant and the main Fuel Enrichment Plant, but that the damage to electricity infrastructure and the resulting loss of power "may have damaged the centrifuges there." However on 17 June the Agency said that subsequent analysis of satellite imagery indicated direct impacts on the underground enrichment halls. According to the IAEA the level of radioactivity outside the Natanz site has remained unchanged indicating no external radiological impact to the population or the environment from this event.

On 16 June the IAEA said it had not seen damage at the site of the Fordow Fuel Enrichment Plant or at the Arak heavy water reactor, which is under construction. The Agency confirmed that neither the Bushehr nuclear power plant nor the Tehran Research Reactor had been targeted or affected by the attacks.

The IAEA said that four buildings were damaged at the Esfahan nuclear site: the central chemical laboratory, a uranium conversion plant, the Tehran reactor fuel manufacturing plant, and the uranium tetrafluoride to enriched uranium metal processing facility, which was under construction.

On 18 June the IAEA said that it had information that two centrifuge production facilities in Iran, the TESA Karaj workshop and the Tehran Research Center, had been hit. At the Tehran site, the Agency said that one building was hit where advanced centrifuge rotors were manufactured and tested, and that at Karaj, two buildings were destroyed where different centrifuge components were manufactured. Both sites were previously under IAEA monitoring and verification activities as part of the JCPOA.

On 19 June Israeli jets bombed the under construction Arak heavy water reactor. The IAEA said that "It was not operational and contained no nuclear material, so no radiological effects." The Agency said that the nearby Heavy Water Production Plant had also been hit.

On 21 June the IAEA said that several buildings at the large nuclear complex in Esfahan had been hit, including: a centrifuge manufacturing workshop, a natural and depleted uranium metal production facility which had not yet begun operations, a fuel rod production facility, a building with low-enriched uranium pellet production as well as a laboratory and nuclear material storage, another laboratory building, a workshop handling contaminated equipment and an office building with no nuclear material.

Overnight on 21/22 June the United States conducted a series of aerial attacks on the sites of Fordow, Natanz and Esfahan. The IAEA said that there had been extensive additional damage at the sprawling Esfahan site. The Agency said that the extent of damage to the Fordow uranium enrichment facility – built deep inside a mountain in central Iran – was not immediately possible to assess given its underground location and the penetrating nature of the bombs used.

The head of the IAEA later said the damage had been substantial and had the effect of bringing Iran's enrichment programme virtually to a halt.

## American-Israeli joint military operation, March 2026

After the 2025 strikes, Iran continued to facilitate IAEA access to unaffected sites, but did not provide reports or access to those that had been hit. Neither did Iran provide access for the IAEA to verify that uranium enrichment was not taking place. These were violations of Iran's obligations under the NPT. In addition, Iran did not provide access for the IAEA to verify that reprocessing and activities related to heavy water were not taking place, as required by resolutions of the UN Security Council and Board.

On 1 March, the US and Israeli militaries began a large-scale campaign attacking Iran's political leadership, military and security forces as well as specific capabilities to manufacture missiles, drones and other weapons. The White House said the operation was in part intended to deny Iran all paths to a nuclear weapon, which it cited as an "imminent threat".

On 2 March 2026, the head of the IAEA told its board of governors: "We have now not had access to Iran's previously declared inventories of low-enriched uranium and high-enriched uranium (over 20%) for more than eight months, making their verification – according to standard safeguards practice – long overdue.

Consequently, the Agency cannot provide assurances in relation to the non-diversion of declared nuclear material from peaceful activities at affected facilities." At that time, the IAEA had no indication that nuclear facilities had been hit.

On 3 March, the IAEA confirmed damage to the entrance buildings to the underground Fuel Enrichment Plant (FEP) at Natanz. In media reports, satellite imagery showed three buildings destroyed.

On 3 March, Rosatom confirmed the precautionary shutdown of Bushehr 1 and suspended all construction work at the Bushehr site. Approximately 639 Russian staff remained onsite, with around 150 having been evacuated. On 11 March Rosatom confirmed that it had evacuated a further 150.

On 4 March, the IAEA said damage to two buildings near the Isfahan site was visible via satellite imagery.

On 18 March, Iran informed the IAEA that a projectile had hit the premises of the Bushehr nuclear power plant, destroying a structure about 350 metres from the reactor. The IAEA confirmed there was no damage to the reactor and no injuries to staff. Rosatom said about 480 of its employees remained at the site and that it was preparing a third stage of evacuation.

## Uranium resources and mining

In the early 1980s, Iran purchased 450 tonnes of uranium (531 t $U_3O_8$) from South Africa. Some 366 t of this was subsequently converted to $UF_6$ at Esfahan. This is the main, and practically the only, material being used in Iran's enrichment plants.

Iran has very small reported uranium resources, all in the <$130/kgU category: 9900 tU as identified resources in situ, including 4300 tU as reasonably assured recoverable; and 4100 tU as inferred resources recoverable (2022 'Red Book'). The ore is reported to have significant levels of molybdenum and other impurities which create difficulties in enrichment. AEOI is responsible for uranium exploration, mining and treatment. Most exploration is in central Iran.

The only mining and milling so far is at Gachin/Gchine, near the port of Bandar Abbas on the Persian Gulf, in the province of Hormozgan. The ore is in surficial salt plugs, accessed by open pit. Total resources here are quoted at 84 tU at 0.068%. The Bandar Abbas Uranium Production Plant (BUPP) began production from Gachin ore in 2006, and operations continue, to produce about 21 tU/yr

with acid leaching. This is delivered to the conversion plant at Isfahan.

An underground mine was developed at Saghand in the central desert region of Yazd province over 1999-2003, and production from the metasomatite ore commenced there in 2017. In 2014, an open pit mine was being developed as mine #2, expected to provide about 30% of output. Resources of 500 tU at 0.0385% U are quoted. The associated Ardakan mill about 75 km west of Saghand is expected to produce 50 tU/yr from the higher-grade ore (average 0.05%) with acid leaching, while lower grade material (0.01 to 0.03%) will be heap leached at the site. The AEOI said that 2017 mine production was expected to be 40 tonnes U. The first uranium oxide production from the mill was announced in April 2018, and sent to the Isfahan conversion plant. A second consignment of 30 tonnes was in January 2019.

Early in 2017, the AEOI announced a proposal to buy 950 tU from Kazakhstan over three years, with 650 tU being delivered in the first two years. The remaining 300 tU would arrive in the third year, be converted to $UF_6$, and sold back to Kazakhstan. The transaction needed approval from the P5+1 group under the 2015 JCPOA. In addition to this, AEOI said that it had received 149 tU from Russia as part of the JCPOA. In April 2017 the AEOI said it had bought 360 tU since July 2015. It also said that Bushehr 1&2 and two other reactors to be built over the next ten years would require 600 tU per year.

In February 2023 the AEOI announced that work had begun at the Narigan Mining and Industrial Complex in Yazd province, the country's largest uranium-molybdenum mine. It said that the site holds 650 tU and 4600 t of molybdenum.

## Organization, regulation & safety

The main organizations involved in the Iranian nuclear industry are tabulated below.

| Organization | Acronym | Est. | Responsibilities |
|---|---|---|---|
| Iran Power Generation and Transmission Company | TAVANIR | 1970 | Involved with policy. |
| Atomic Energy Organization of Iran | AEOI | 1974 | The lead authority, responsible for the establishment of regulations for nuclear and |

| Organization | Acronym | Est. | Responsibilities |
|---|---|---|---|
| | | | radiation safety (under a 1989 act), licensing facilities, and supervising. |
| Iran Nuclear Regulatory Authority | INRA | 1975 | Responsible for regulation and safety, monitoring, legal compliance and radioactive waste management. It is under the AEOI and maintains a close relationship with its Russian counterpart, Rostechnadzor. |
| Nuclear Science & Technology Research Institute | NSTRI | 2002 | Assumed responsibility over AEOI's research role. |
| Nuclear Power Production & Development Company of Iran | NPPD | 2004 | Responsible for Bushehr. |

# Non-proliferation

| 1970s |
|---|
| Iran joined the Nuclear Non-Proliferation Treaty (NPT) in 1970 and concluded its safeguards agreement with the IAEA in 1974. It has signed the Additional Protocol to this safeguards agreement but has not ratified it; voluntary implementation, which began in 2003, was suspended in February 2021. |
| **2002** |
| Iran attracted world attention when some previously undeclared nuclear facilities became the subject of IAEA inquiry. On investigation, the IAEA found inconsistencies in Iran's declarations to the Agency and raised questions as to whether Iran was in violation of its safeguards agreement, as a signatory of the NPT. |
| **2003** |
| All Iran's facilities, except the Arak heavy water plant and a Kalaye plant, were under IAEA safeguards as of mid-2003. Details are in the Director-General's reports to the IAEA Board on the IAEA website.<br>An IAEA report in November showed that Iran had, in a series of contraventions of its safeguards agreement over 22 years, systematically concealed its development of key techniques which are capable of use for nuclear weapons. In particular, the enrichment of uranium and plutonium separation from used fuel were carried out on a laboratory scale. Iran admitted to the activities, but said they were trivial. |
| **2005** |
| In August, the IAEA Board called upon Iran to suspend work associated with uranium enrichment. |
| **2006** |

In March the IAEA referred the issue to the UN Security Council, which required that “Iran shall without further delay suspend . . . all enrichment-related and reprocessing activities.” However, Iran has not backed off from its activities in developing uranium enrichment.

## 2007

On 24 March, the UN Security Council unanimously adopted a resolution imposing further sanctions on Iran and reaffirming that Iran must take the steps required by the IAEA Board, notably to suspend its uranium enrichment activities. The IAEA reported in May that Iran had ceased providing information required under the Additional Protocol.
The IAEA stated clearly in November and since, that unless the Additional Protocol was ratified and in place it is not possible for the Agency to establish that undeclared nuclear materials and activities are absent. Its "knowledge about Iran's current nuclear program is diminishing". Meanwhile enrichment continues, the existence of the underground and undeclared Qom plant has come to light, and hence a third UN Security Council resolution appeared likely.
The Iran situation revived wider concerns about which countries should develop facilities with high proliferation significance – such as enrichment and reprocessing, even under safeguards if there is no evident economic rationale. At some point in the future, such a country could give three months notice of withdrawal from the NPT and reconfigure its facilities for weapons production. The USA asserted that Iran had been in fact developing just such a breakout capability.

## 2010

This contention was supported in February when the government ordered the AEOI to commence enriching Iranian uranium to 19.75% for the Tehran Research Reactor (TRR), thereby significantly closing the gap between its normal low-enriched material and weapons-grade uranium. The 1950 kg of low-enriched uranium (< 5%) moved to PFEP would be enough for vastly more 19.75% enriched uranium than the TRR could conceivably use.

## 2011

In August the AEOI confirmed that Iran had more 20% LEU than it needed for the Tehran research reactor, and that “security measures required that the sensitive part of the facilities would be transferred to underground buildings” – evidently Fordow.

## 2012

Iran continues to deny the IAEA’s requests for access to the alleged high explosive testing site related to nuclear weaponization experiments at Parchin. Iran’s extensive activities there – including significant asphalting – seriously undermined the Agency’s ability to conduct effective verification.
This loomed as a factor to block implementation of the 2013 Geneva agreement, and remains a significant issue in “the clarification of past and present outstanding issues” regarding suspected nuclear weapons activities under the 2015 Joint Comprehensive Plan of Action.

## 2013

On 11 November a new Joint Statement on a Framework for Cooperation was signed with the IAEA whereby “Iran and the IAEA will cooperate further with respect to verification activities to be undertaken by the IAEA to resolve all present and past issues”. It addressed several of IAEA’s secondary concerns (other than the main centrifuge enrichment program and the heavy water reactor) but set a three-month target for six measures. The agreement was essentially a test of possible enhanced cooperation on the main issues and to set up a negotiating process. This initiative was in parallel with the P5+1 negotiations.

| 2014 |
| --- |
| Iran agreed with the IAEA on safeguard measures for the IR-40 heavy water reactor at Arak and pledged to finalize these by August.<br>In September, the IAEA reported that Iran was not meeting its obligations under the Framework. |
| **2022** |
| The UN report in December highlighted that no progress had been made on the implementation of the 2015 Security Council resolution, with Iran still enriching "worrying quantities" of uranium. The IAEA reported that Iran intended to install new centrifuges at one of its fuel enrichment plants, whilst also enriching uranium up to 60% at another plant. It believed that the country then had a total enriched uranium stockpile of more than 18 times the allowable amount under the JCPOA and that its ability to effectively monitor the country's nuclear facilities was compromised. |
| **2023** |
| In February 2023 France, Germany, the UK and the USA called on Iran to comply with all its legally-binding international obligations under its safeguards agreement. The following month, Iran agreed to enhance its cooperation with the IAEA, including the reinstallation of monitoring cameras at enrichment sites. |
| **2025** |
| On 12 June the IAEA's board of governors formally declared Iran in breach of its non-proliferation obligations for the first time in 20 years. |

## November 2013 Geneva Agreement: Interim Joint Plan of Action

An agreement to curb Iran's evident progress towards nuclear weapons capability was struck on 24 November 2013 between Iran and the foreign ministers of China, France, Germany, Russia, UK, and USA (P5+1 – the five permanent members of the UN Security Council plus Germany) and a senior EU representative. It linked closely to the IAEA Joint Statement on a Framework for Cooperation signed two weeks earlier, and over the next 16 months proved effective in rolling back Iran's nuclear program for the first time in a decade, applying innovative inspections measures, allowing only modest sanctions relief and keeping substantial pressure on Iran.

The initial steps of the Interim Joint Plan of Action covered a term of six months, renewable by mutual consent. For that period, Iran undertook not to enrich uranium to over 5% U-235, nor to make "any further advances" of activities at its Natanz and Fordow enrichment plants or the Arak heavy water reactor. It would not install any further centrifuges beyond the 18,500 then, nor operate

the 1000 or so advanced centrifuges among these. It also undertook to dilute half of its "working stock" of 20%-enriched uranium to no more than 5%, with the remainder retained as oxide for fabrication of fuel for the Tehran Research Reactor (TRR). Enhanced monitoring activities would include wider access for IAEA inspectors and provision of information to the IAEA. In return, the countries undertook to lift various US and EU sanctions on sectors including petrochemical exports, gold and precious metals and promised that no new nuclear-related sanctions would be imposed by either the UN Security Council or the EU over the six-month period covered by the first step, which commenced on 20 January 2014.

The Geneva interim plan aimed to resolve two key issues before a comprehensive deal could be finalized. First, all IAEA concerns about Iran's past, and possibly ongoing, work on nuclear weapons and other alleged military nuclear activities must be satisfied – Iran had stalled on this for years, notably in blocking IAEA access to the Parchin site. Secondly, Iran needed to address satisfactorily all provisions of UN Security Council resolutions, if not by suspending its centrifuge program as earlier demanded, then taking enough tangible steps to alleviate international concerns about this aspect of its nuclear program and also the Arak heavy water reactor.

According to the action plan, the parties aimed to conclude negotiating and begin implementing a long-term "comprehensive solution" within a year of the adoption of the Geneva agreement. The comprehensive solution would involve a mutually defined enrichment program with "practical limits and transparency measures" to ensure that Iran's nuclear program remains peaceful; "Iran reaffirms that under no circumstances will Iran ever seek or develop any nuclear weapons," the action plan states. The ultimate objective is the lifting of all nuclear-related sanctions against Iran, and the country's nuclear program being treated in the same way as that in any other non-nuclear weapon state that is party to the nuclear non-proliferation treaty (NPT).

The IAEA report on 20 March 2014 detailed Iran's progress. The report on 23 May 2014 said that Iran had:

- Not enriched any uranium above 5% at any declared facility.
- Downblended 100 kg of near 20% enriched $UF_6$ and converted the rest to oxide (essentially leaving none as $UF_6$, but with now about 230 kg as oxide).

- Not installed any further centrifuges at the Natanz FEP or Fordow.
- Not progressed work on the Arak IR-40 reactor or made fuel for it.
- Provided access to the Arak heavy water plant.
- Continued R&D at the Natanz PFEP.

The 20 July 2014 IAEA report confirmed these data for the near 20% enriched material, and that 1505 kg of ca. 5% enriched $UF_6$ had been converted to $UO_2$. As 24 November 2014 approached, the Institute for Science and International Security (ISIS) said:

*In order to avoid a bad deal, the P5+1 must hold strong on achieving an agreement that limits Iran's nuclear program to a reasonable civilian capability, significantly increases the timelines for breakout to nuclear weapons, and introduces enhanced verification that goes beyond the IAEA's Additional Protocol. A sound deal will also require Iran to verifiably address the IAEA's concerns about its past and possibly on-going work on nuclear weapons, which means Iran must address those concerns in a concrete manner before a deal is finalized or any relief of economic or financial sanctions occurs.*

## Framework for Joint Comprehensive Plan of Action, Lausanne, April 2015

On 2 April 2015 a framework agreement was struck by the P5+1 group and Iran, taking forward the November 2013 interim Joint Plan of Action and forming the foundation upon which the final text of the Joint Comprehensive Plan of Action could be written by the end of June. It reflects the significant progress made in discussions between the P5+1, the European Union and Iran, though it confers some legitimacy to Iran's enrichment program. Important implementation details remain subject to negotiation, and the agreement was described as 'fragile' pending finalization. Negotiators worked to conclude the Joint Comprehensive Plan of Action (JCPOA) based on the framework so that it could be signed and sealed at the end of June. It plausibly extends Iran's 'breakout capability' from an estimated 2-3 months currently to at least a year, and does so for a decade or more. Beyond that, Iran will be bound by its long-term enrichment and R&D plan presented to the P5+1 group.

It was agreed that Iran would reduce its installed enrichment centrifuges from about 19,000 to 6,104, only 5,060 of which will be in use for ten years, enriching to no more than 3.76%. All of them will be first-generation IR-1 centrifuges – none of its more advanced models can be used for ten years, and the 1000 IR-2M centrifuges at Natanz will be removed and stored under IAEA monitoring for those ten years. Iran would then be allowed a gradual increase in (enrichment) capacity between the 10th and 13th years with the introduction of advanced IR-2 and IR-4 centrifuges. Meanwhile, any R&D into more efficient designs, IR-4, IR-5, IR-6 and IR-8, will have to be based on a plan submitted to the IAEA. For the IR-8, only single machines can be tested for the first 8.5 years after the implementation of the agreement, after which testing of cascades of up to 30 machines may begin. Fordow will cease all enrichment and be turned into a physics research centre. It will not produce or house any fissile material for at least 15 years and most of its centrifuges will be removed and placed under IAEA monitoring. The LEU stockpile will be reduced from 10,000 kg to 300 kg of 3.67% LEU for 15 years.

The heavy water reactor at Arak will be redesigned and rebuilt according to a design agreed by the P5+1, and its original core will be destroyed or removed from the country. All used fuel from the reactor will be shipped abroad, indefinitely. No other heavy water reactor will be built for 15 years, and surplus heavy water will be sold abroad.

Although there is no reference to military sites such as Parchin, under the terms of framework agreement, an intensive inspection regime is created, which will remain in place indefinitely. Inspectors from the IAEA should be able to access any facility, declared or otherwise, and Iran will be required to grant access to the IAEA to investigate 'suspicious sites' or allegations of covert facilities. Iran has agreed to ratify and implement the Additional Protocol under the NPT, ensuring ongoing transparency and IAEA access. Robust inspection of Iran's uranium supply chain will be undertaken for 25 years.

As a result, there would be phased relief from American and EU sanctions, as long as Iran complies, though US nuclear sanctions would be retained on the books so as to allow response in the event of significant non-compliance. The main sanctions imposed by the UN Security Council resolutions will be lifted once key concerns are addressed. However, core provisions in the UN Security Council resolutions dealing with transfer of sensitive technologies and activities would be re-established by a new Security Council resolution which endorses the Joint Comprehensive Plan of Action.

## Joint Comprehensive Plan of Action, Vienna, July 2015

In mid-July 2015, the Joint Comprehensive Plan of Action with Iran was signed, after protracted negotiations. Iran agreed that over the next 15 years it will not enrich uranium above 3.67% and will reduce its stockpile of low-enriched uranium (over 200kg of it near 20% enriched) from 9000 to 300 kg of enriched uranium. Uranium research and development activities will only take place at Natanz, with much reduced number of centrifuges, while no enrichment will be carried out at the underground Fordow site. At Natanz, the number of installed centrifuges will be reduced from 19,500 to 6,100, only 5,000 of which will be spinning. All of them will be first-generation types: none of its more advanced models can be used for at least 10 years and R&D into more efficient designs will have to be based on a plan submitted to the IAEA. In addition, Iran has agreed indefinitely not to build any new heavy water reactors or stockpile heavy water and redesign the Arak reactor, with the original core being removed and destroyed. All used fuel will be shipped out of the country.

A separate agreement with the IAEA set out a path for "the clarification of past and present outstanding issues" regarding suspected nuclear weapons activities. Explanations and interactions were scheduled to mid-October so that the IAEA Board could receive a full report in December. Once the IAEA confirms that Iran has complied with its obligations under the international agreement, economic sanctions will progressively be lifted. The IAEA welcomed Iran's decision to implement the Additional Protocol to its Comprehensive Safeguards Agreement with the IAEA, allowing the intrusive monitoring required under the Joint Comprehensive Plan of Action (JCPOA).

The Carnegie Endowment commented positively:

*The agreement demonstrates the viability of the rules-based nonproliferation regime created by the Nuclear Non-Proliferation Treaty (NPT) and including especially the IAEA safeguards system, notwithstanding the lacunae and imperfections of this regime. Indeed, the JCPOA buttresses the NPT. Whereas states may withdraw from the NPT and, in principle, then seek nuclear weapons, in the JCPOA Iran has committed not to ever seek nuclear weapons under any circumstances. And whereas the NPT does not include specific restrictions on activities that could contribute to the design and development of a nuclear explosive device, the JCPOA does.*

On 13 October, the Iranian parliament approved a bill on the implementation of the JCPOA. Following this, 18 October 2015 was Adoption Day for JCPOA, and all participants started to prepare for implementing their respective commitments, including lifting sanctions once the IAEA had verified that Iran had completed all its steps.

China, the USA and Iran also released a joint statement of intent on 18 October on the future steps of modernization of the Arak reactor as specified under the JCPOA. This includes the removal of its existing calandria and rendering it inoperable. The China Atomic Energy Authority is responsible for undertaking this work.

**Implementation of the Joint Comprehensive Plan of Action**

In mid-November 2015 the IAEA reported that a total of 4,112 IR-1 centrifuges and related infrastructure had been removed from service at the Natanz fuel enrichment plant along with 160 IR-2m centrifuges and related infrastructure. These centrifuges and other equipment are being stored at the site, under IAEA verification and monitoring. At the same time a total of 258 IR-1 centrifuges and related infrastructure were removed from the Fordow plant.

ISIS calculates from the IAEA report that in November 2015, Iran had a total inventory of low-enriched uranium comprising 8.3 tonnes of 3.5% uranium hexafluoride (5.57 tU) and 4.3 tonnes mostly as oxide (2.9 tU) at the Isfahan EUPP. This stock had increased by 460 kg since August. In addition, there was the 228 kg of near 20% enriched uranium in various forms.

In December 2015, Iran shipped more than 11 tonnes of various low-enriched uranium materials to Russia, in accordance with the JCPOA. The shipment included all of Iran's nuclear material enriched to near 20% that was not already in the form of fabricated fuel plates for the Tehran Research Reactor.* The IAEA now must verify that Iran's enriched uranium stockpile is 300 kg or less at under 3.67% U-235. In exchange for the enriched uranium, Iran received an equivalent amount of natural uranium, 137 tonnes, most originating in Kazakhstan.

* In May 2015 Iran had 228 kg U as 19.75% enriched uranium, 61.5 kg as oxide powder, 44.9 kg as TRR fuel, and 121.2 kg as scrap, waste, or in-process, and none as hexafluoride. The total of 228 kg U was the same in November, and 51.5 kg of this was in TRR fuel assemblies, plus a couple of kg in target irradiation plates. The remaining 175 kg would need to be exported under the JCPOA. This is half of what had been produced before some was downblended.

The core of the Arak heavy water reactor was removed in January 2016 and concrete was poured into the calandria.

Implementation Day came in mid-January 2016, when the IAEA verified that Iran had completed all of its nuclear commitments, the report of which was submitted to the IAEA Board and the UN Security Council. The effect of these actions is to increase Iran's "breakout time" to obtain enough nuclear material for a weapon to one year, up from less than 90 days before the JCPOA. Iran will now start to implement provisionally the Additional Protocol to its IAEA safeguards agreement, which together with other measures under the JCPOA, will increase the agency's ability to monitor nuclear activities in Iran and verify that they are peaceful.

On 8 May 2018, the USA withdrew from the deal. In January 2021, the IAEA reported that Iran had resumed enriching uranium to 20% purity at its underground Fordow plant.

See also information page on Nuclear Proliferation Case Studies.

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